



UNIVERSITY OF
PLYMOUTH

PUSL3190 Computing Project

Project Interim Report

Multi-Point Spatial-Temporal Verification Framework
for

Fraud-Resistant Vehicle Access Control in
RF-Congested Industrial Environments

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Chapter 01: Introduction

1.1 Introduction

Over the past decade, industrial access control has become heavily dependent on radio-frequency identification (RFID) technology. These systems perform reliably in clean, low-interference environments, but they face serious operational challenges when placed in settings where electromagnetic noise is a constant factor. Colombo Dockyard PLC is one such setting. The facility is a major ship repair and heavy engineering operation in Sri Lanka, handling over 200 vessels per year with a workforce of more than 1,000 employees. Currently, the dockyard runs a UHF RFID-based vehicle attendance system. The system's core purpose is not simply to control vehicle access; it is used to determine which employees have brought their personal registered vehicles to the facility on a given day, which in turn determines their eligibility for a monthly fuel allowance.

The RFID system has deteriorated significantly due to two compounding problems. First, the Colombo Harbor Authority operates its own high-powered RFID infrastructure immediately adjacent to the dockyard, using the same 860 to 960 MHz frequency band. Radio signals do not stop at physical fences, so there is continuous signal interference between the two sites. Second, the dockyard itself is filled with large metallic structures: ship hulls, steel gantries, and stacked shipping containers. These surfaces scatter and reflect radio waves in unpredictable patterns, which causes both read failures on valid tags and phantom reads on unrelated signals. The combination of these two factors has made the system unreliable to the point where it can no longer be trusted for administrative decisions.

This project proposes a complete architectural shift away from radio-frequency verification. The replacement system is built around optical detection using Automatic License Plate Recognition (ALPR), powered by a deep learning pipeline. The distinguishing feature of the proposed framework is a spatial-temporal correlation engine that models vehicle movement across the facility as a directed graph. Rather than verifying a token that can be removed and carried separately, the system verifies the physical vehicle itself and checks whether its movement between checkpoints is physically possible. This approach eliminates the fundamental weaknesses of the existing setup.

1.2 Problem Definition

The problems with the current system are not limited to hardware malfunctions. They reflect deeper architectural flaws that make the system vulnerable to both technical failures and deliberate abuse. Three separate issues contribute to the overall failure.

1. **RF Physics Failure:** The dockyard environment is saturated with metallic surfaces that act as reflectors for UHF radio signals. When the RFID readers broadcast their interrogation signals, those signals bounce off ship hulls and containers, creating interference patterns throughout the site. In some locations, overlapping signals cancel each other out, leaving dead zones where valid tags go unread even when the vehicle is directly in front of the reader. In other locations, the signals from the Harbor Authority's system reach the dockyard readers, causing false positive reads that have nothing to do with actual vehicle arrivals. These are not software bugs; they are properties of electromagnetic wave behaviour in metallic spaces, and no amount of reconfiguration will resolve them in this environment.
2. **Architectural Vulnerability (Token vs. Asset Verification):** The most significant design flaw in the current system is that it authenticates a credential rather than the asset the credential is meant to represent. Each eligible employee has a passive RFID windshield tag linked to their employee record. The system treats a successful tag read as proof that the corresponding vehicle has entered the facility. This assumption is exploitable. An employee can remove the RFID tag from their windshield, arrive at the facility by public transport, and present the tag directly to the reader as if they were scanning it from inside a vehicle. Gate staff often allow this without challenge, partly because RFID read failures are so common that manual tag scans appear to be a normal workaround for a faulty system. This pass-back fraud directly generates the financial losses reported in internal audits.
3. **Stateless Verification Gap:** Replacing the RFID readers with commercial ALPR cameras would solve the interference problem but would not address fraud by itself. Most commercial ALPR systems process each detection as a standalone event, with no memory of where a vehicle was previously seen. If a cloned plate is used, or if an employee passes their key card to a colleague who enters through a different gate moments later, a stateless system would log both events as valid without flagging the physical contradiction. A vehicle cannot appear at Gate 1 and then at Gate 3 twelve seconds later if the minimum driving time between those gates is three minutes.

Detecting this kind of temporal impossibility requires the system to maintain state and apply physical constraints to detection sequences.

The financial impact of these combined failures has been documented through internal audits. The Finance Department has recorded a consistent 15 to 20 percent discrepancy between RFID-logged vehicle entries and the actual count of vehicles in the parking area. This gap translates to an estimated annual overpayment of 2.7 to 3.6 million LKR in fuel allowances. On top of the direct financial cost, the HR and Finance departments together spend roughly 35 hours per month resolving allowance disputes, which represents a significant drain on administrative resources.

1.3 Project Objectives

The project targets a set of specific, measurable outcomes that address each identified failure layer.

- Design and implement an edge-processed dual-camera optical detection system using the YOLOv8-Small architecture, trained specifically on Sri Lankan license plates captured under adverse maritime conditions including salt spray, monsoon rain, and variable lighting.
- Develop a character recognition pipeline using a Convolutional Recurrent Neural Network (CRNN), supported by a Weighted Homoglyph Matching algorithm that corrects OCR errors common to local plate fonts, such as misreading '8' as 'B', '0' as 'O', or '1' as 'I'.
- Build a Spatial-Temporal Correlation (STC) Engine that models the facility's access points and internal road network as a directed graph, capable of detecting fraud by identifying vehicle movement sequences that violate physical travel time constraints.
- Reduce the false-positive verification rate from approximately 20 percent to below 2 percent, restoring confidence in the automated attendance records.
- Provide a tamper-evident audit log using PostgreSQL that stores cropped plate images, recognition confidence scores, and timestamped access decisions for each transaction, targeting a reduction of at least 90 percent in the time spent on allowance dispute investigations.

Chapter 02: System Analysis

2.1 Facts Gathering Techniques

The project opened with a structured investigation phase designed to produce a factual, evidence-based understanding of how the existing system fails and what operational conditions the replacement must handle. Three complementary methods were used.

- **Semi-Structured Interviews:** Interviews were conducted with staff from three departments who interact with the vehicle attendance system daily.
 - Security personnel described the operational strain during peak shift change periods. When more than 200 vehicles arrive within a 30-minute window and the RFID system fails to read a tag, a guard must step out of the booth, locate the vehicle registration on a printed list, and manually authorise entry. This process takes between 30 and 45 seconds per vehicle. During busy periods, the backlog grows faster than it can be cleared, and guards are eventually forced to wave vehicles through without verification in order to prevent traffic from spilling onto the public road.
 - Finance and HR staff highlighted the administrative burden created by disputed allowance claims. Because the system produces no photographic evidence, employees who claim a read failure occurred cannot be disproven. The default response is to approve the disputed allowance, which reinforces the incentive for fraud.
- **On-Site Observation:** A physical survey of Gate 1 and Gate 3 confirmed the cross-site interference problem directly. During the observation session, the dockyard's RFID readers produced audible read alerts on multiple occasions when Harbor Authority heavy vehicles passed on the other side of the perimeter fence, well outside the dockyard boundary. This was documented as direct evidence of frequency contamination between the two adjacent sites.
- **Document and Data Analysis:** Access logs, internal audit records, and operational reports covering late 2024 and early 2025 were reviewed in detail. These documents established the quantitative baseline for the project: a confirmed 15 to 20 percent fraud discrepancy rate and a monthly financial leakage of between 225,000 and 300,000 LKR in unwarranted fuel allowances.

2.2 Existing System

The current vehicle attendance system is the third iteration the facility has attempted. The first system, based on Electronic Toll Collection technology, was retired due to vendor dependency and excessive maintenance downtime. The second system used a basic computer vision approach but produced too many false detections and could not handle older Sri Lankan plate formats or motorcycles. The current generation relies entirely on long-range UHF RFID.

In normal operation, eligible employees receive a passive UHF RFID windshield tag that is linked to their employee record in a central MS SQL Server database. As a vehicle pulls up to Gate 1 or Gate 3, a directional overhead antenna emits a signal that powers the passive tag through inductive coupling. The tag responds by broadcasting its 96-bit EPC identifier, which the reader forwards over TCP/IP to the application server. If the identifier matches an active employee record, the server logs a timestamped entry event, triggers the relay to raise the boom barrier, and sets a boolean flag in the HR database marking the employee as eligible for that day's fuel allowance.

2.3 Drawbacks of the Existing System

The investigation identified three categories of failure in the existing RFID implementation.

- **Electromagnetic Interference Susceptibility:** Operating in the 860 to 960 MHz band, the system is vulnerable to both destructive and constructive interference from nearby metallic structures and the adjacent Harbor Authority system. Destructive interference creates read nulls where valid tags are not detected even at close range. Constructive interference extends the reader's effective range unpredictably, pulling in signals from unintended sources and logging phantom entries.
- **Authentication Decoupling:** The system authenticates a credential object rather than the vehicle itself. An employee can separate the RFID tag from their vehicle, bring only the tag to the gate, and trigger a valid entry event without a vehicle ever entering the facility. The frequency of legitimate system failures provides cover for this behaviour, as manual tag presentations cannot easily be distinguished from compensation for a malfunctioning reader.
- **No Trajectory Validation:** The database records each detection as an isolated event: a gate ID, a tag ID, and a timestamp. There is no mechanism to correlate sequential events for the same tag. A tag appearing at Gate 1 at 08:00:00 and again at Gate 3 at

08:00:15 is logged as two valid entries. The system has no concept of whether such a movement is physically achievable, so it cannot detect duplicate scans, credential sharing, or cloned tag attacks based on timing alone.

Chapter 03: Requirements Specification

3.1 Functional Requirements

The following functional requirements define what the system must be able to do in order to replace the existing infrastructure and address its identified weaknesses.

- **FR-01 (Optical Plate Localisation):** The system must autonomously detect and crop license plate regions from a live 2K video stream. Each localisation must complete within 100 milliseconds per frame so that the pipeline can operate at real-time speeds without causing gate delays.
- **FR-02 (Sequence Extraction and Normalisation):** The system must extract the alphanumeric character sequence from a cropped plate image. It must handle Sri Lankan plate syntax specifically, including stripping vertically printed provincial codes such as 'WP' or 'SP', and reformatting legacy plate series before performing database lookups.
- **FR-03 (Weighted Fuzzy Validation):** The system must compare extracted plate text against the registered vehicle database using a Modified Levenshtein Edit Distance algorithm configured with an optical confusion penalty matrix. Substitutions that correspond to known camera blur artefacts must carry a low penalty, while unrelated character substitutions must carry a full penalty, preventing lookalike plates from gaining access.
- **FR-04 (Spatial-Temporal Validation Engine):** The system must maintain a record of each vehicle's last confirmed location and the time of that confirmation. Each new detection must be evaluated against the physical minimum travel time between the previous gate and the current gate. If the elapsed time is less than the minimum physically possible travel time, the system must block access and generate a Spatial Impossibility Alert.
- **FR-05 (Cryptographic Audit Logging):** Every access transaction must produce a structured log record containing the cropped plate image, the raw OCR output, the normalised plate string, the detection timestamp, the gate identifier, the recognition confidence score, and the final access decision. These records must be stored in a tamper-evident format suitable for Finance Department review.

3.2 Non-Functional Requirements

The following non-functional requirements govern the performance, reliability, and practicality of the deployed system.

- **NFR-01 (Detection Accuracy):** The system must achieve a minimum combined detection and recognition accuracy of 85 percent under the full range of maritime environmental conditions present at the dockyard, including direct sun glare on wet metallic surfaces, heavy rainfall obscuring plate characters, and progressive salt spray accumulation on camera lenses.
- **NFR-02 (Processing Latency):** The end-to-end pipeline, measured from frame capture through model inference, fuzzy matching, spatial-temporal validation, and database logging, must complete within 1.5 seconds at the 95th percentile. Exceeding this threshold risks creating vehicle queues at the gate that spill onto adjacent roads.
- **NFR-03 (Hardware Resilience):** All externally mounted components, including camera housings and LED display units, must carry a minimum IP67 ingress protection rating. The corrosive salt-laden atmosphere at a maritime facility significantly accelerates hardware degradation, making this rating a minimum viable standard.
- **NFR-04 (Cost Constraint):** The total hardware and compute cost per gate must remain below 100,000 LKR. This threshold ensures the solution is substantially cheaper than enterprise ALPR platforms and is achievable within the facility's operational budget.

3.3 Hardware and Software Requirements

Hardware Requirements

- **Edge Compute Node:** Local processing is required to avoid cloud API latency and maintain operation during network outages. Each gate requires a workstation with a modern CPU (Intel Core i7 or equivalent) and an NVIDIA GPU with at least 6 GB of VRAM (RTX 3060 or higher) to run the YOLOv8 and CRNN models concurrently at the required speed.
- **IP Cameras:** Two cameras are required per gate, positioned to capture both the front and rear of each vehicle. This dual setup handles partial occlusion caused by closely following vehicles and ensures that motorcycles, which carry plates only at the rear, are reliably captured. Both cameras must support 2K resolution and Wide Dynamic Range (WDR) processing to handle backlight and sun glare.

- **LED Display Panel:** A P10 outdoor LED matrix panel (standard 32x16 module configuration) provides immediate visual feedback to drivers. Messages such as 'ACCESS GRANTED' or 'INVALID PLATE' are displayed as soon as the verification result is produced.

Software Requirements

- **Operating System:** Ubuntu 22.04 LTS is the target deployment environment. Windows 10 or 11 Professional is used for initial development and testing.
- **Programming Language:** Python 3.9 or later is used throughout the project. The language's broad support for machine learning libraries and computer vision frameworks makes it the natural choice for this type of application.
- **AI Frameworks:** The Ultralytics library provides the YOLOv8 implementation for plate detection. PyTorch 2.0 is used to train and run the custom CRNN character recognition model.
- **Vision and Optimisation Libraries:** OpenCV handles image preprocessing tasks including format conversion and contrast enhancement. NVIDIA TensorRT is used to quantise the trained PyTorch models from FP32 to INT8 or FP16 precision, significantly reducing inference time on the target GPU.
- **Database:** PostgreSQL is selected over lightweight alternatives such as SQLite because of its Multi-Version Concurrency Control (MVCC) capability. MVCC allows multiple camera processing threads to write audit log entries simultaneously without causing locks or data corruption, which is a practical requirement during high-traffic periods.

Chapter 04: Feasibility Study

4.1 Operational Feasibility

From an operational standpoint, the proposed system directly removes the most disruptive element of the current workflow: the manual fallback procedure required when RFID reads fail. At present, each failed read costs between 30 and 45 seconds of guard time and contributes to vehicle queues that can exceed the available gate area during shift changes. The proposed system processes each vehicle in under 1.5 seconds without requiring any guard intervention for standard cases.

A key design consideration is the handling of edge cases where the optical system cannot produce a reliable result. Rather than failing hard, the system presents an override option on the security dashboard. This keeps a human decision-maker in the loop for ambiguous situations such as severely damaged plates or unusual lighting conditions, while automating all routine verifications. This design is expected to improve adoption among security staff, since it reduces workload without removing their ability to intervene.

4.2 Economic Feasibility

Commercial enterprise ALPR systems from vendors such as Vigilant Solutions, Genetec, or Hikvision's analytics platform typically cost between USD 10,000 and USD 50,000 per gate, with additional recurring fees for software licences and cloud processing services. These costs place them well outside the facility's budget and make them economically difficult to justify for a single-site deployment.

By building the system on open-source deep learning frameworks and consumer-grade GPU hardware, the estimated capital cost per gate for the proposed solution falls between 85,000 and 100,000 LKR. Given that the Finance Department's audits show a monthly fuel allowance overpayment of between 225,000 and 300,000 LKR attributable to the current system's failures, the two-gate installation would recover its full cost within approximately three months of operation. The return on investment profile is, by any standard measure, highly favourable.

4.3 Technical Feasibility

The project team has the required background in Python development, computer vision, and relational database management. The primary technical challenge is not the general approach

but the specific deployment context. Deep learning models trained on standard Western ALPR datasets perform poorly on Sri Lankan plates, particularly under the visual degradation conditions present at the dockyard. This risk is addressed directly by constructing a custom training dataset sourced from the actual deployment site, with deliberate inclusion of difficult cases such as mud-covered plates, faded characters, and oblique camera angles.

Processing latency is the other major technical risk. The 1.5-second requirement is tight for a multi-stage pipeline running two neural network models. This risk is managed through NVIDIA TensorRT model quantisation, which converts the FP32 model weights to INT8 or FP16 representations. This reduces model size and increases throughput on the target RTX 3060 GPU substantially, and preliminary benchmarks suggest the latency target is achievable with the selected hardware configuration.

Chapter 05: System Architecture

5.1 Use Case Diagram

Three actor roles interact with the system within the dockyard environment.

- **Driver:** Drivers are passive participants in the verification process. They pull up to a gate camera, stop at the barrier, and read the LED display to confirm whether their vehicle has been granted access. No action is required from the driver unless access is denied.
- **Security Officer:** Security staff monitor the live verification dashboard, which shows incoming detections, recognised plate strings, confidence scores, and any fraud alerts raised by the spatial-temporal engine. Officers have access to a manual override control for cases where the automated result needs to be reviewed or corrected.
- **System Administrator:** Administrators manage the backend of the system. Their responsibilities include maintaining the employee and vehicle registry, updating the directed graph parameters that define physical travel time limits between gates, and exporting the signed audit logs requested by the Finance Department for monthly reconciliation.

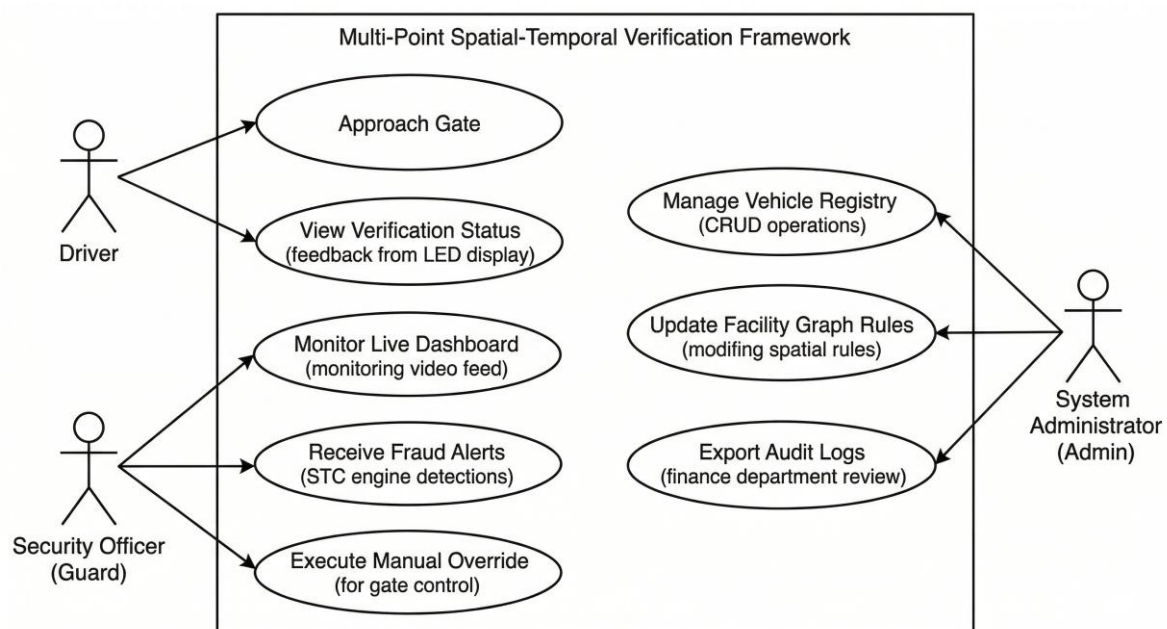


Figure 5.1 - Use Case Diagram of the Multi-Point Spatial-Temporal Verification Framework

5.2 Class Diagram of Proposed System

The software design separates the perception layer from the verification logic layer. The five primary classes and their relationships are described below.

- **CameraStream:** This class uses a background thread to continuously pull frames from the camera's RTSP feed into a shared buffer. The main processing thread always reads the most recent frame from this buffer rather than waiting for the capture operation, which prevents processing lag from accumulating during high-load periods.
- **PlatePerception:** This class wraps the YOLOv8 detection model and the CRNN recognition model into a single interface. It accepts a raw image frame, returns a cropped plate image, and provides the extracted character string.
- **SpatialTemporalVerifier:** This class implements the core fraud detection logic. It holds an in-memory dictionary mapping each recognised plate string to a VehicleState object that records the last confirmed gate and timestamp for that vehicle.
- **FacilityGraph:** This class represents the physical layout of the dockyard as a directed graph. Each node corresponds to a gate camera, and each edge carries a minimum travel time value in seconds derived from the physical road distance and the facility's internal speed limit.
- **FraudAlert:** This class is a structured data object that is created only when the SpatialTemporalVerifier determines that a vehicle's transition between two gates violates the physical travel time constraint defined in the FacilityGraph.

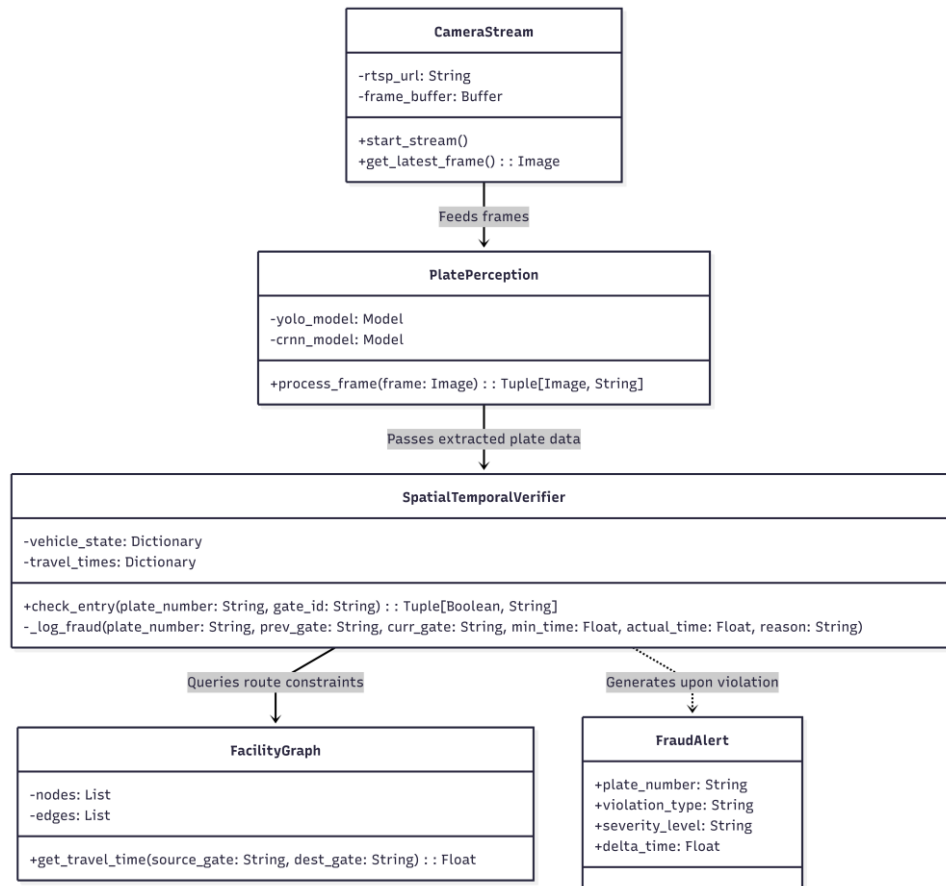


Figure 5.2 - Class Diagram of the Proposed System

5.3 Entity-Relationship Diagram

The database schema is designed to support non-repudiable audit records. The four core entities and their relationships are as follows.

- Entities:** Employee (EmpID, Name, Department); Vehicle (PlateNo, VehicleType, EmpID); AccessLog (LogID, PlateNo, GateID, Timestamp, OCR_Confidence, OverrideStatus); FraudEvent (EventID, PlateNo, ViolationType, SeverityLevel, DeltaTime).
- Relationships:** One Employee may register one or more Vehicles (1:N). One Vehicle may generate many AccessLog entries over time (1:N). Where an AccessLog entry corresponds to a detected physical impossibility, it generates one FraudEvent record (1:1).

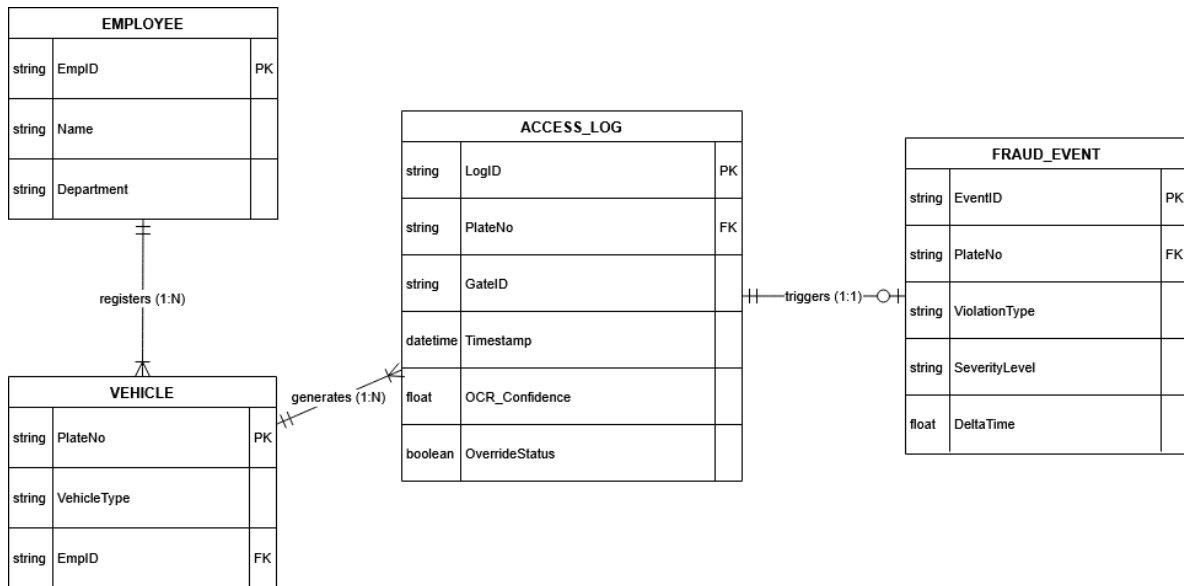


Figure 5.3 - Entity-Relationship Diagram of the database schema

5.4 High-Level Architecture

The complete verification pipeline is divided into three functional layers.

4. **Perception Layer:** Both gate cameras feed frames simultaneously into the edge compute node. YOLOv8 processes each frame to isolate and crop the plate region. The CRNN model then reads the character sequence from the cropped image.
5. **Normalisation and Matching Layer:** The raw OCR string passes through a regular expression normaliser that removes provincial prefix codes and standardises the format. The cleaned string is then matched against the employee vehicle database using the Weighted Homoglyph Matching algorithm, which applies optical confusion penalties to produce a match confidence score.
6. **Spatial-Temporal Validation Layer:** The matched plate identity and the precise detection timestamp are passed to the STC Engine. The engine checks the vehicle's previous location and applies the FacilityGraph's travel time constraints. If the movement is physically plausible, the system sends an open signal to the gate relay and commits a 'GRANTED' log entry. If the movement is impossible, access is denied, a FraudEvent is created, and the security dashboard displays an alert.

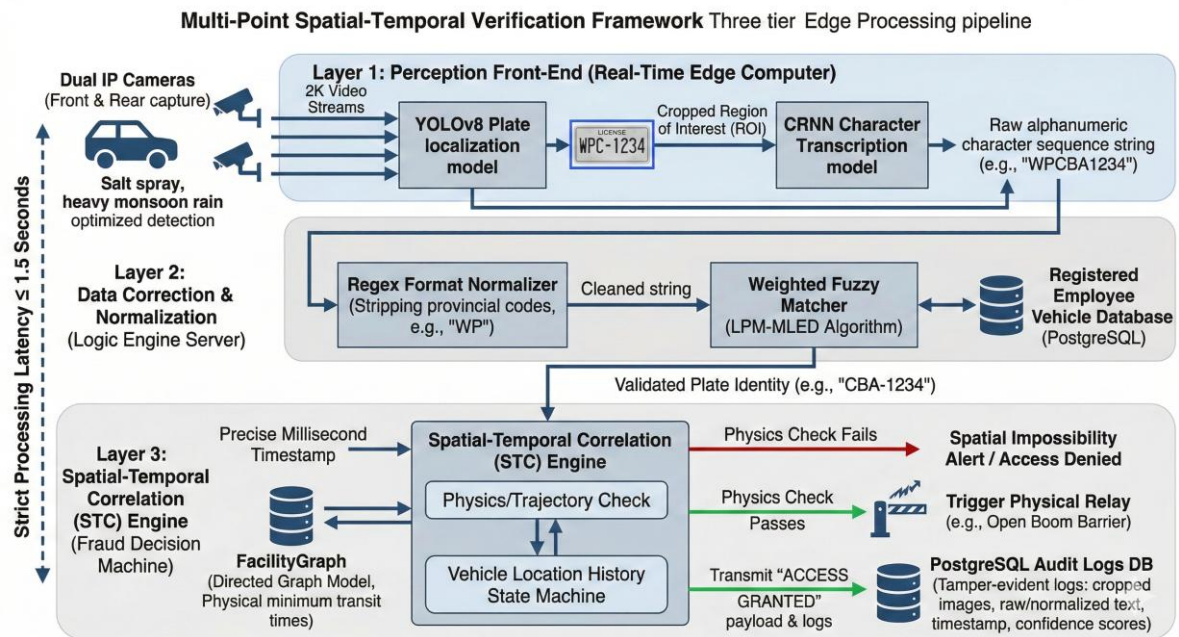


Figure 5.4 - High-Level Three-Tier Edge Processing Pipeline Architecture

Chapter 06: Development Tools and Technologies

6.1 Development Methodology

The project uses an Agile iterative methodology built around two-week development sprints. This choice was driven by the unpredictable nature of training deep learning models for a specific deployment environment. A waterfall plan would commit to accuracy targets upfront, but whether those targets are achievable depends entirely on the quality and volume of training data collected from the actual site. Iterative sprints allow the team to adapt based on real test results rather than theoretical projections.

Each sprint follows the same cycle: data collection, model training, inference testing, and hyperparameter adjustment. If a sprint reveals that the recognition model consistently misclassifies a particular character, the next sprint prioritises collecting more examples of that character and refining the confusion penalty matrix accordingly. This feedback loop is what allows the model to converge on adequate performance for local conditions rather than generic ones.

6.2 Programming Languages and Tools

- **Python 3.9+:** Python is the primary development language. Its ecosystem for deep learning, image processing, and data management is broader and more actively maintained than any alternative, making it the standard choice for applied computer vision work.
- **Ultralytics YOLOv8:** YOLOv8 is used for the plate detection stage. Compared to earlier YOLO versions, the v8 architecture uses an anchor-free detection head, which improves performance on small objects at varying angles and distances. This characteristic is particularly relevant for license plates captured from overhead cameras at entry gates.
- **PyTorch 2.0:** PyTorch provides the deep learning backend for the CRNN character recognition model. Its dynamic computation graph makes it easier to inspect model behaviour during training and to modify the architecture between sprint iterations.
- **PostgreSQL:** PostgreSQL is the database engine for the audit logging backend. Its MVCC architecture allows multiple concurrent write operations from different camera

threads without requiring explicit locking, which is critical during peak arrival periods when several detections may occur within the same second.

6.3 Third-Party Libraries

- **OpenCV (cv2):** OpenCV handles all image preprocessing tasks that occur before frames are passed to the neural network models. The library provides efficient implementations of operations such as colour space conversion, contrast enhancement, and geometric normalisation that would be impractical to write from scratch.
- **EasyOCR:** EasyOCR was used in the early prototyping phase to establish a performance baseline. The custom-trained CRNN model is benchmarked against EasyOCR results throughout development to confirm that the localised training effort produces measurable gains over a general-purpose solution.

6.4 Algorithms

Two original algorithmic contributions form the intellectual core of this project.

7. **Modified Levenshtein Edit Distance (LPM-MLED):** Standard edit distance algorithms assign a uniform substitution cost of 1.0 regardless of which characters are involved. This does not reflect optical reality: a camera blurred image of the digit '8' is far more likely to be misread as 'B' than as 'W'. The LPM-MLED algorithm addresses this by applying a penalty matrix in which optically similar character pairs carry a discounted cost (0.1), while unrelated pairs retain the full cost (1.0). A match is accepted if the total computed distance is 0.5 or less. This configuration reduces false rejections caused by camera noise while maintaining meaningful resistance to lookalike plate attacks.
8. **Directed Graph Fraud Detection:** The dockyard is modelled as a directed graph $G = (V, E)$ where vertices represent gate camera nodes and edges represent physical road connections. Each edge is weighted with the minimum travel time in seconds, calculated from the road distance and the facility's internal speed limit. Each time a vehicle is detected at a gate, the engine retrieves the vehicle's previous gate and the elapsed time since that detection. If the elapsed time is less than the edge weight for that route, the system raises a fraud alert. Elapsed times below 0.5 seconds trigger a Clone Attack classification, since no physical movement could account for such an

interval; longer violations that still fall short of the minimum travel time are classified as Speeding Violations, which may indicate credential sharing.

Chapter 07: Implementation Progress

7.1 Development Environment Setup

The development environment has been fully configured and tested. The primary workstation runs Windows 11 with an NVIDIA RTX 3060 GPU (6 GB VRAM), an Intel Core i7 processor, and 16 GB of DDR4 RAM. NVIDIA CUDA 11.8 and the corresponding cuDNN libraries are installed to enable GPU-accelerated tensor operations for model training and inference.

Dataset management is handled through the Roboflow platform, which provides version control for annotated image sets and supports collaborative labelling. Roboflow's built-in augmentation pipeline is used to artificially extend the training dataset by applying transformations such as simulated blur, rotation, and brightness variation. Source code is managed through a standard Git repository with branch-level isolation between feature development and the stable training baseline.

7.2 Implemented Features

Progress has been made across all three pipeline layers. The system has moved from design to a functional, demonstrable prototype.

- **Dataset Construction:** A localised dataset of 327 fully annotated images has been assembled, representing approximately 65 percent of the 500-image target. The dataset was curated with a focus on challenging cases that a generic dataset would not include: plates covered partially in mud, characters faded by salt corrosion, plates photographed under direct equatorial sunlight, and the stacked provincial code format specific to Sri Lankan plates.
- **Layer 1 (Plate Detection):** The YOLOv8 Small model has been trained for 50 epochs on the custom dataset. Validation results show a detection precision of 87.3 percent and a recall of 84.1 percent. Inference runs at approximately 12 milliseconds per frame on the RTX 3060, which is well within the budget available for the detection stage of the 1.5-second pipeline.
- **Layer 3 (Spatial-Temporal Engine):** The complete STC Engine has been written, integrated with the FacilityGraph representation, and unit-tested against a range of synthetic detection sequences including normal entries, clone attack patterns, and credential-sharing scenarios.

- **Tabletop Demonstration Prototype:** A scaled physical prototype has been built to validate the STC logic in a controllable setting. The prototype uses two USB webcams, a printed dockyard gate layout, and 1:18 scale die-cast vehicle models fitted with printed license plates scaled to produce sufficient pixel density for the detection models. The prototype allows the team to physically simulate and verify fraud detection scenarios for the academic viva, independent of the real camera deployment.



Figure 7.1 - YOLOv8 bounding box detection on a Sri Lankan license plate (tabletop prototype, Gate 1)



Figure 7.2 - Physical tabletop prototype showing dual webcam setup and 1:18 scale model vehicles

7.3 Code Snippets

The following two extracts demonstrate the key technical implementations completed to date.

Snippet 1: Spatial-Temporal Correlation Engine (Layer 3)

This function implements the physics-based fraud detection logic. It retrieves the vehicle's previous gate and timestamp, queries the FacilityGraph for the minimum required travel time on that route, and classifies the event based on the elapsed time.

```
# src/spatial.py
def check_entry(self, plate_number, gate_id):
    """
    Core logic for Multi-Point Spatial-Temporal Verification.
    Detects Speeding, Clone Attacks, and Path Violations.
    """
    now = datetime.now()

    # New vehicle entering for the first time
    if plate_number not in self.vehicle_state:
        self.vehicle_state[plate_number] = {'gate': gate_id, 'timestamp': now}
        return (True, "System Entry")

    prev = self.vehicle_state[plate_number]
    route_key = (prev['gate'], gate_id)
    min_required_time = self.travel_times.get(route_key)
    actual_time = (now - prev['timestamp']).total_seconds()

    if actual_time < min_required_time:
        if actual_time < 0.5:
            reason = f"CLONE ATTACK: {prev['gate']}->{gate_id} in
{actual_time:.2f}s"
        else:
            reason = f"SPEEDING VIOLATION: {prev['gate']}->{gate_id} in
{actual_time:.2f}s"
        self._log_fraud(plate_number, prev['gate'], gate_id,
                        min_required_time, actual_time, reason)
        return (False, reason)

    self.vehicle_state[plate_number] = {'gate': gate_id, 'timestamp': now}
    return (True, f"Valid: {actual_time:.1f}s")
```

Snippet 2: Maritime Image Preprocessing (CLAHE)

This function applies Contrast Limited Adaptive Histogram Equalisation to improve plate legibility in hazy or low-contrast conditions before passing the image to the recognition pipeline.

```
# vision_pipeline.py
import cv2

def enhance_plate_contrast(cropped_plate_img):
    # Convert to LAB colour space to isolate the lightness channel
    lab = cv2.cvtColor(cropped_plate_img, cv2.COLOR_BGR2LAB)
    l_channel, a, b = cv2.split(lab)

    # Apply CLAHE with localised 8x8 tile grids to avoid global noise
    # amplification
    clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8, 8))
    cl = clahe.apply(l_channel)

    # Reconstruct image in BGR for downstream processing
    merged = cv2.merge((cl, a, b))
    return cv2.cvtColor(merged, cv2.COLOR_LAB2BGR)
```

7.4 Challenges and Solutions

- **Challenge 1: Maritime Visual Degradation:** Test footage collected from the actual gate cameras showed that salt spray deposits on the lens surface produce a diffuse haze layer that dramatically reduces plate contrast. Standard OCR engines, including the EasyOCR baseline, failed on the majority of frames collected under these conditions.
 - Solution: A preprocessing step using OpenCV's CLAHE algorithm was introduced into the pipeline. CLAHE divides the image into small local tiles (8x8 pixels in this implementation) and equalises contrast within each tile independently. This restores legible contrast in the plate region without amplifying the background haze, which global histogram equalisation would do. Accuracy under hazy conditions improved measurably after this change.
- **Challenge 2: Provincial Code Parsing:** Sri Lankan number plates often include a two-letter provincial code printed vertically on the left side of the plate. The OCR engine frequently incorporated these characters into the main plate string, producing outputs like 'WPCAB1234' where the registered entry was 'CAB-1234'. These mismatches caused the database lookup to fail entirely.
 - Solution: A normalisation layer was built using Python's regular expression library. It detects known provincial prefix patterns at the start of the recognised string, strips them, and reinserts the standard hyphen separator before the numeric portion. The plate string is then reformatted to the database's expected schema before the matching query is made.
- **Challenge 3: Prototype Scale Limitation:** The first prototype iteration used 1:64 scale die-cast vehicles. The license plates on these models measure approximately 4 millimetres in width. Standard 720p webcams cannot resolve text at that scale from a practical camera mounting distance, and the detection model returned no useful bounding boxes.
 - Solution: The prototype was rebuilt using 1:18 scale vehicles. Custom license plates were printed at twice the scale that would be accurate for the model, producing physical plate widths of approximately 18 millimetres. This provided enough pixel density for the detection and recognition models to operate successfully during viva demonstrations.

7.5 Current Limitations

The following issues have been identified and are scheduled for resolution before the final submission.

- **Motorcycle Detection:** The current YOLOv8 model struggles with rear-mounted motorcycle plates. These plates are smaller and more variable in aspect ratio than car plates, and the existing training dataset does not include sufficient examples. A dedicated detection class for motorcycle plates will be added in the next training run.
- **Nighttime and Infrared Operation:** The current pipeline operates exclusively in the visible light spectrum. Detection accuracy degrades substantially in low-light conditions, particularly when combined with heavy rain. A full production deployment would require IR illuminator cameras and IR-optimised plate detection logic. This capability is outside the scope of the current project phase and is noted here as a deployment prerequisite.

Chapter 08: Discussion

Summary of the Report

This interim report has documented the full development arc of a vehicle access control system designed to replace the failing RFID infrastructure at Colombo Dockyard PLC. The investigation phase established that the existing system's problems stem from two sources: the physical limitations of radio-frequency detection in a metallic maritime environment, and a fundamental design flaw that authenticates a removable token rather than the vehicle it is supposed to represent. Both issues are well-documented and are beyond the reach of configuration-level fixes.

The proposed solution addresses these problems through three pipeline layers: optical plate detection using YOLOv8, character normalisation and fuzzy matching using a custom penalty matrix, and physics-based fraud detection using a directed graph model of the facility. At this stage of development, the YOLOv8 model achieves 87.3 percent detection precision on the custom dataset, the STC Engine has been tested successfully against all planned fraud scenarios, and a working physical prototype exists that can demonstrate the system's fraud detection logic in a controlled setting.

Changes from the Original Proposal

The project's scope has shifted meaningfully since the initial PID submission. The original proposal framed the work primarily as an attendance automation system, with the main deliverable being a mechanism for calculating fuel allowances more reliably than the current RFID setup.

During early supervision discussions, it became clear that this framing did not provide sufficient academic scope for a final-year engineering project. The focus was repositioned towards security engineering and physics-based fraud detection. The critical addition was the Layer 3 Spatial-Temporal Engine, which introduces an original algorithmic contribution by applying directed graph modelling and physical constraint validation to access control. This change elevated the project from a reimplementing of existing ALPR technology to a research-oriented contribution with novel elements, specifically the use of movement physics as a fraud detection mechanism in a real industrial access control context.

Future Work

The remaining work before the April 2026 submission is structured around four tasks.

- **Dataset Expansion and Model Retraining (Weeks 18 to 20):** The training dataset will be extended from 327 to at least 500 annotated images. The expanded set will include additional motorcycle plate examples and more low-light captures. The CRNN character recognition model will be retrained on the expanded dataset with the aim of pushing recognition accuracy above the 90 percent threshold.
- **Hardware Integration (Weeks 20 to 22):** The software backend will be connected to physical IoT relay modules to simulate actual barrier gate operation. The LED display driver will also be integrated, allowing the full end-to-end flow from detection to physical gate actuation and visual feedback to be demonstrated.
- **Stress Testing and Latency Profiling (Weeks 24 to 28):** The system will be subjected to simulated burst traffic loads equivalent to 200 vehicle detections within a 30-minute window. The goal is to confirm that both the PostgreSQL write throughput and the inference pipeline can sustain these loads without the 95th percentile latency exceeding 1.5 seconds.
- **Final Thesis Documentation:** The complete project thesis will document the architecture design decisions, the mathematical formulation of both algorithms, and a comparative performance evaluation against the EasyOCR baseline and the existing RFID system's reported metrics.

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